

Document 1

What Linux Is

Linux is an open-source operating system used on desktops, servers, mobile devices, and embedded systems. It is known for stability, security, and flexibility. With a variety of distributions available, each offers different tools and environments. Linux is widely used by developers, system administrators, cybersecurity professionals, and anyone who prefers a customizable system.

Popular Linux Distributions

- Debian: A stable, community-driven distribution known for reliability.
- Ubuntu: Beginner-friendly and Debian based, with strong community support.
- Fedora: A distribution backed by Red Hat.
- Arch Linux: A rolling-release distro for advanced users who want full control.

Components of the Linux OS

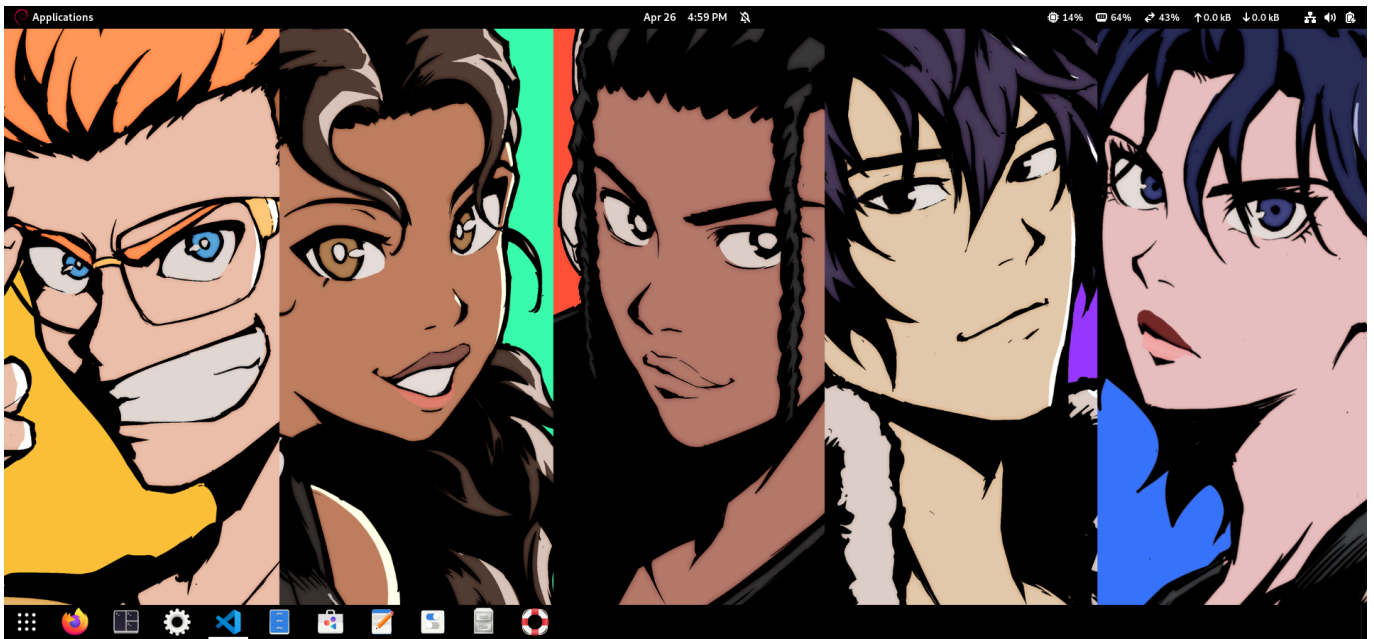
Component	Description
Bootloader	Manages the boot process and loads the operating system.
Kernel	The core of the OS; manages CPU, memory, and hardware.
Init System	Starts user space and controls system services (daemons).
Daemons	Background services for printing, sound, scheduling, networking, and more.
Graphical Server	Displays graphics on the monitor; often called the X server or X11.
Desktop Environment	The user interface (GNOME, KDE, XFCE, etc.) with built-in tools and apps.
Applications	Software installed on top of the desktop environment for daily use.

Steps to Install a Linux Distro

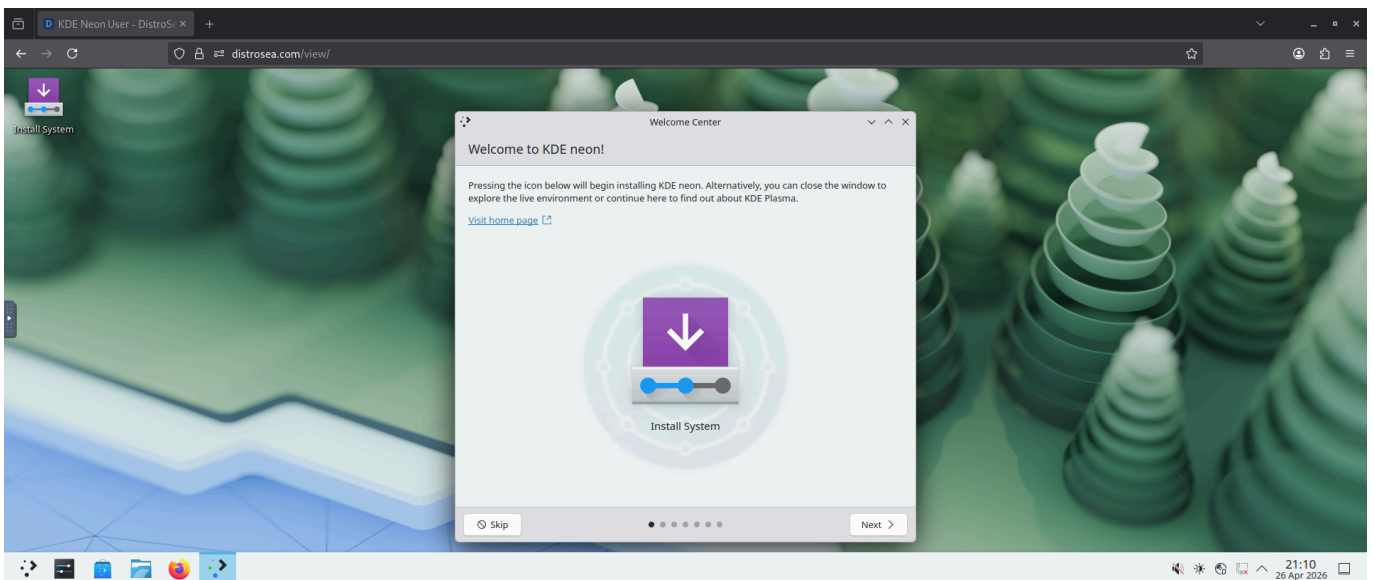
1. Download an ISO
2. Create a bootable USB or VM
3. Boot into the installer
4. Follow installation prompts
5. Reboot into your new system

Desktop Environments

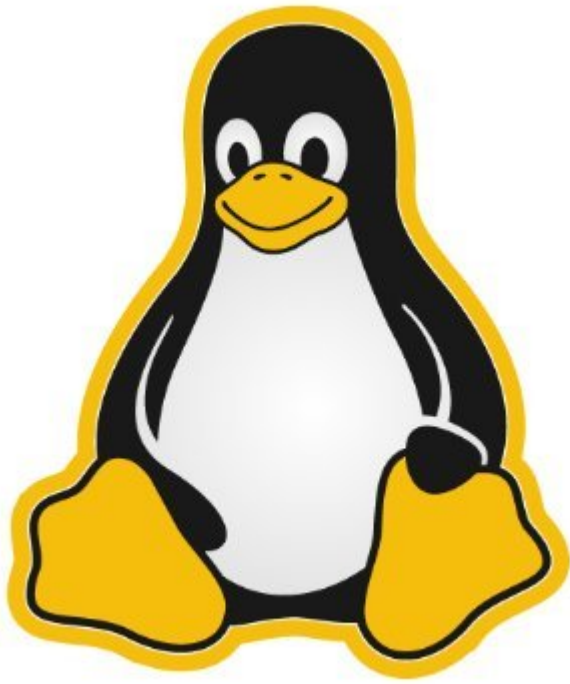
GNOME:



KDE Neon:



Linux Mascot



Linux Distribution Logos



Document 2

Getting Started With Python

Python is a beginner-friendly programming language used for automation, data work, scripting, and general software development. It focuses on readability and simple syntax, making it easy to learn for new programmers and powerful enough for advanced users.

Learn the Basics

Python code is written in plain text files ending in .py. The interpreter runs each line from top to bottom. Indentation is meaningful and defines code blocks.

Hello, World!

A simple program that prints text to the screen:

- `print("Hello, World!")`

Variables and Types

Python creates variables when you assign values to them. Common types include integers, floats, strings, and booleans.

type	example
integer	<code>x = 5</code>
float	<code>pi = 3.14</code>
string	<code>"hello"</code>
boolean	<code>True</code>

Lists

Lists store ordered collections of items. `colors = ["red", "green", "blue"]`

Basic Operators

Python supports arithmetic (+ - * / %) and comparison operators (== != > < >= <=) used in expressions.

String Formatting

You can insert variables into strings using f-strings:

```
name = "Dan" print(f"Hello, {name}")
```

Basic String Operations

Basic String Operations allow you too inspect and modify strings. `len(word)` - counts how many characters are in a string `text[-1]` - Negative indexing accesses characters from the end of a string. `[-1]` being the last character.

Conditions

Conditional statements let your program make decisions.

```
if x > 10: print("Large") else: print("Small")
```

Loops

Loops repeat actions.

```
for i in range(3): print(i)
```

Document 3

Table 1

ID	Name	Email	Investments
231	Albert Master	albert.master@gmail.com	Bonds
210	Alfred Alan	aalan@gmail.com	Stocks
256	Alison Smart	asmart@biztalk.com	Residential Property
211	Ally Emery	allye@easymail.com	Stocks
248	Andrew Phipps	andyp@mycorp.com	Stocks
234	Andy Mitchel	andym@hotmail.com	Stocks
226	Angus Robins	arobins@robins.com	Bonds
241	Ann Milan	ann_milan@iinet.com	Residential Property
225	Ben Bessel	benb@hotmail.com	Stocks
235	Benson Romanov	benr@albert.net	Bonds

Table 2

Letter Grade	Percentage
A	90-100%
B	80-89.99%
C	70-79.99%
D	60-69.99%
F	<60%

Table 3

Part	Name
	Graphics Card

Part	Name
	Hard Drive
	Motherboard
	RAM